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Display device and electronic device

Abstract

A second display region has a lower pixel density than a first display region. A third metal layer is located upper than a first metal layer and a second metal layer. The occupancy of the third metal layer in the second display region is lower than the occupancy in the first display region. In a pixel unit, the first metal layer includes a first electrode region to control an amount of electric current in a driving transistor, the second metal layer includes a second electrode region and a third electrode region to supply current to the driving transistor, and the third metal layer includes a main region to form a capacitor included in a capacitive element to store a control voltage for the driving transistor, and an island region surrounded by the main region with a gap and interconnected with a lower electrode region by a via region.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This non-provisional application claims priority under 35 U.S.C. § 119(a) on Patent Application No. 2021-120823 filed in Japan on Jul. 21, 2021, the entire content of which is hereby incorporated by reference.

BACKGROUND

(2) This disclosure relates to a display device.

(3) An organic light-emitting diode (OLED) element is a current-driven self-light-emitting element and therefore, does not need a backlight. In addition to this, the OLED element has advantages for achievement of low power consumption, wide viewing angle, and high contrast ratio; it is expected to contribute to development of flat panel display devices.

(4) The display region of an OLED display device can include a region having a pixel density different from the pixel density of the other region. For example, some portable terminals, such as smartphones and tablet computers, include a photosensitive element like a camera for taking a picture under the display region. For the camera to receive external light, the camera is disposed under a region having a lower pixel density than the surroundings.

SUMMARY

(5) An aspect of this disclosure is a display device including: a first display region configured to display an image; a second display region configured to display an image, the second display region having a lower pixel density than the first display region; a first metal layer; a second metal layer located upper than the first metal layer; and a third metal layer located upper than the first metal layer and the second metal layer. Occupancy of the third metal layer in the second display region is lower than occupancy of the third metal layer in the first display region. The first display region includes a plurality of first pixel units and each of the plurality of first pixel units includes: a light-emitting element including an upper electrode region, a lower electrode region, and a light-emitting layer located between the upper electrode region and the lower electrode region; and a driving transistor configured to control light emission of the light-emitting element. The first metal layer, the second metal layer, and the third metal layer are located lower than the lower electrode region. In each of the plurality of first pixel units, the first metal layer includes a first electrode region configured to control an amount of electric current in a channel of the driving transistor. In each of the plurality of first pixel units, the second metal layer includes a second electrode region and a third electrode region configured to supply electric current to the channel of the driving transistor. In each of the plurality of first pixel units, the third metal layer includes: a main region configured to be supplied with a power supply potential and form a capacitor included in a first capacitive element with the second metal layer, the first capacitive element being configured to store a voltage to control the driving transistor; and an island region separated from the main region and surrounded by the main region with a gap, the island region being interconnected with the lower electrode region by a via region.

(6) An aspect of this disclosure is a display device including: a first display region configured to display an image; a second display region configured to display an image, the second display region having a lower pixel density than the first display region; a first metal layer; a second metal layer located upper than the first metal layer; and a third metal layer located upper than the first metal layer and the second metal layer. A pattern of the third metal layer in the second display region is different from a pattern of the third metal layer in the first display region. The first display region includes a plurality of first pixel units and each of the plurality of first pixel units includes: a light-emitting element including an upper electrode region, a lower electrode region, and a light-emitting layer located between the upper electrode region and the lower electrode region; and a driving transistor configured to control light emission of the light-emitting element. The first metal layer, the second metal layer, and the third metal layer are located lower than the lower electrode region. In each of the plurality of first pixel units, the first metal layer includes a first electrode region configured to control an amount of electric current in a channel of the driving transistor. In each of the plurality of first pixel units, the second metal layer includes a second electrode region and a third electrode region configured to supply electric current to the channel of the driving transistor. In each of the plurality of first pixel units, the third metal layer includes: a main region configured to be supplied with a power supply potential and form a capacitor included in a first

capacitive element with the second metal layer, the first capacitive element being configured to store a voltage to control the driving transistor; and an island region separated from the main region and surrounded by the main region with a gap, the island region being interconnected with the lower electrode region by a via region.
(7) It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of this disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** schematically illustrates a configuration example of an OLED display device;
(2) FIG. **2** illustrates a configuration example of a pixel circuit and control signals therefor in an embodiment of this specification;
(3) FIG. **3** is a plan diagram schematically illustrating a device structure of one pixel unit;
(4) FIG. **4** schematically illustrates the cross-sectional structure along the section line IV-IV′ in FIG. **3**;
(5) FIG. **5**A illustrates a pattern of a layer included in the layered structure of a TFT substrate;
(6) FIG. **5**B illustrates a pattern of another layer included in the layered structure of the TFT substrate;
(7) FIG. **5**C illustrates a pattern of still another layer included in the layered structure of the TFT substrate;
(8) FIG. **5**D illustrates patterns of some layers included in the layered structure of the TFT substrate;
(9) FIG. **5**E illustrates a pattern of still another layer included in the layered structure of the TFT substrate;
(10) FIG. **5**F illustrates a pattern of still another layer included in the layered structure of the TFT substrate;
(11) FIG. **5**G illustrates a pattern of still another layer included in the layered structure of the TFT substrate;
(12) FIG. **6** schematically illustrates a configuration example of a display region;
(13) FIG. **7** schematically illustrates a layout of control lines on the TFT substrate;
(14) FIG. **8**A illustrates a first metal layer pattern, an intermediate conductive layer pattern, and a third metal layer pattern in another structural example of a pixel unit including a third metal layer;
(15) FIG. **8**B illustrates the intermediate conductive layer pattern and the third metal layer pattern in the structural example illustrated in FIG. **8**A;
(16) FIG. **8**C illustrates a second metal layer pattern in addition to the layered structure illustrated in FIG. **8**A;
(17) FIG. **9** is a cross-sectional diagram schematically illustrating the cross-sectional structure of the other structural example of a pixel unit; and
(18) FIG. **10** is a cross-sectional diagram schematically illustrating the cross-sectional structure of a low-density region including a third metal layer having another example of pattern.
EMBODIMENTS
(19) Hereinafter, embodiments of this disclosure will be described with reference to the accompanying drawings. It should be noted that the embodiments are merely examples to implement this disclosure and not to limit the technical scope of this disclosure.
(20) In the following description, a pixel is the smallest unit in the display region and an element for emitting light of a single color. It is also referred to as subpixel. A set of pixels of different colors (red, blue, and green, for example) constitute an element for displaying one mixed-color dot. This element can be referred to as main pixel. When the element for emitting light of a single color

needs to be distinguished from the element for emitting light of a mixed color for clarity of description, the former is referred to as subpixel and the latter as main pixel. The features of this description are applicable to monochrome display devices whose display region is composed of monochrome pixels.

(21) A configuration example of a display device is described. The display region of the display device includes a region (also referred to as low-density region or low-resolution region) having a relatively low pixel density and a region (also referred to as normal region or normal-resolution region) having a relatively high pixel density. The display region can include a plurality of low-density regions where the pixel density is lower than the normal region and these low-density regions can have different pixel densities. The light-emitting element of a pixel in the examples described in the following is a current-driven element, such as organic light-emitting diode (OLED) element.

(22) Raising the density of pixel circuits to attain higher resolution may cause some problems as follows. One of them is that the power lines are thinned (narrowed) and have higher resistance, which causes non-uniformity of the anode power supply voltage (IR drop) within the plane. Another one is that the storage capacitor gets closer to lines and a crosstalk occurs because of capacitive coupling, so that the storage capacitor cannot retain an accurate potential (meaning that desired light cannot be obtained from a light-emitting element). Still another one is that, in the case where the display device is a flexible display device, thin metal lines provided on an insulating layer or a polysilicon layer are easily broken when the substrate is bent.

(23) A pixel unit of a display device in an embodiment of this specification includes a first metal layer including gate electrode regions, a second metal layer including source/drain electrode regions, and moreover, a third metal layer located upper than these layers. A gate electrode region is a first electrode region for controlling the amount of electric current in the channel of a transistor. A source/drain electrode region is a second electrode region or a third electrode region for supplying the electric current to the channel of the transistor.

(24) The third metal layer is supplied with a power supply potential for the light-emitting element and forms a storage capacitor for the driving field-effect transistor with the second metal layer. The display region includes a normal region and a low-density region, between which the third metal layer has different patterns. This configuration reduces the IR drop of the power supply potential and further, provides characteristics suitable for each display region to the pixel units therein.

(25) Configuration of Display Device

(26) An overall configuration of the display device in an embodiment of this specification is described with reference to FIG. **1**. The elements in the drawings may be exaggerated in size or shape for clear understanding of the description. In the following, an organic light-emitting diode (OLED) display device is described as an example of the display device.

(27) FIG. **1** schematically illustrates a configuration example of an OLED display device **10**. The OLED display device **10** includes a thin-film transistor (TFT) substrate **100** on which OLED elements (light-emitting elements) are fabricated and a structural encapsulation unit **150** for encapsulating the OLED elements. In the periphery of a cathode electrode region **114** outer than the display region **125** of the TFT substrate **100**, control circuits, specifically a scanning driver **131**, an emission driver **132**, an electrostatic discharge protection circuit **133**, a driver IC **134**, and a demultiplexer **136**, are provided.

(28) The driver IC **134** is connected to the external devices via flexible printed circuits (FPC) **135**. The scanning driver **131** drives scanning lines on the TFT substrate **100**. The emission driver **132** drives emission control lines to control light emission of pixels. The electrostatic discharge protection circuit **133** saves the elements on the TFT substrate **100** from electrostatic discharge damage. The driver IC **134** is mounted with an anisotropic conductive film (ACF), for example.

(29) The driver IC **134** provides power and control signals including a timing signal to the scanning driver **131** and the emission driver **132** and further, provides power and a data signal to the

demultiplexer **136**. The demultiplexer **136** outputs output of one pin of the driver IC **134** to d data lines (d is an integer greater than 1) in series. The demultiplexer **136** changes the output data line for the data signal from the driver IC **134** d times per scanning period to drive d times as many data lines as output pins of the driver IC **134**.

(30) Pixel Circuit

(31) FIG. **2** illustrates a configuration example of a pixel circuit **200** and control signals therefor in an embodiment of this specification. The pixel circuit **200** is included in an N-th pixel circuit row (N is an integer). The pixel circuit **200** includes six transistors (TFTs) P**1** to P**6** each having a gate, a source, and a drain. All transistors P**1** to P**6** in this example are p-type TFTs.

(32) The transistor P**1** is a driving transistor for controlling the amount of electric current to an OLED element E**1**. The source of the driving transistor P**1** is connected with a power line **241** for transmitting a power supply potential VDD. The driving transistor P**1** controls the amount of electric current to be supplied from the power line **241** to the OLED element E**1** in accordance with a voltage stored in a storage capacitive element C**0**. The storage capacitive element C**0** holds the written voltage throughout one frame period. The cathode of the OLED element E**1** is connected with a power line **204** for transmitting a power supply potential VEE from a cathode power supply.

(33) The storage capacitive element C**0** in the configuration example of FIG. **2** consists of a first capacitive element C**1** and a second capacitive element C**2** connected in series. One end of the storage capacitive element C**0** is supplied with the anode power supply potential VDD and the other end is connected with the gate of the driving transistor P**1**.

(34) The storage capacitive element C**0** consists of the capacitive elements C**1** and C**2** connected in series in between the power line **241** for transmitting the anode power supply potential VDD and the gate of the driving transistor P**1**. An end of the capacitive element C**1** is connected with the power line **241** and the other end of the capacitive element C**1** is connected with an end of the capacitive element C**2**. The other end of the capacitive element C**2** is connected with the gate of the driving transistor P**1**. The source/drain of the transistor P**4** and the source/drain of the transistor P**2** are connected with an intermediate node between the capacitive elements C**1** and C**2**.

(35) The voltage of the storage capacitive element C**0** is a voltage between the gate of the driving transistor P**1** and the VDD power line **241**. The source of the driving transistor P**1** is connected with the VDD power line **241**; the source potential is the anode power supply potential VDD. Accordingly, the storage capacitive element C**0** stores the gate-source voltage of the driving transistor P**1**.

(36) The transistor P**5** is an emission control switching transistor for controlling ON/OFF of supply of driving current to the OLED element E**1** and the resulting light emission of the OLED element E**1**. The source of the transistor P**5** is connected with the drain of the driving transistor P**1**. The transistor P**5** switches ON/OFF the current supply to the OLED element E**1** connected with its drain. The gate of the transistor P**5** is connected with a control signal line **232**A and the transistor P**5** is controlled by the emission control signal Em input from the emission driver **132**. The emission control signal is a selection signal for controlling light emission of the OLED element E**1**.

(37) The transistor P**6** works to supply a reset potential Vrst to the anode of the OLED element E**1**. One end of the source/drain of the transistor P**6** is connected with a power line **242** for transmitting the reset potential Vrst and the other end is connected with the anode of the OLED element E**1**.

(38) The gate of the transistor P**6** is connected with a control signal line **231**A and the transistor P**6** is controlled by a selection signal S**1**. When the transistor P**6** is turned ON by the selection signal S**1** from the scanning driver **131**, the transistor P**6** supplies the reset potential Vrst transmitted by the power line **242** to the anode of the OLED element E**1**. The transistors P**5** and P**6** also supply the reset potential Vrst to the gate of the driving transistor P**1** via the transistor P**3**.

(39) The transistor P**3** is a switching transistor (threshold compensation transistor) for writing a voltage for applying threshold calibration (threshold compensation) to the driving transistor P**1** to the storage capacitive element C**0** and is also a transistor for resetting the gate potential of the

driving transistor P**1**. The source and the drain of the transistor P**3** connect the gate and the drain of the driving transistor P**1**. Accordingly, when the transistor P**3** is ON, the driving transistor P**1** is diode connected.
(40) The transistor P**4** is a switching transistor (threshold compensation transistor) for writing a voltage for applying threshold compensation to the driving transistor P**1** to the storage capacitive element C**0**. The transistor P**4** controls whether to supply a reference potential Vref to the storage capacitive element C**0**. One end of the source/drain of the transistor P**4** is connected with a power line **202** for transmitting the reference potential Vref and the other end is connected with an intermediate node between the capacitive elements C**1** and C**2**. The gate of the transistor P**4** is connected with a control signal line **231**B and the transistor P**4** is controlled by the selection signal S**1** input from the scanning driver **131** to its gate.
(41) The transistors P**3**, P**6**, and P**4** are controlled by the selection signal S**1**. Accordingly, these transistors P**3**, P**6**, and P**4** are turned ON/OFF simultaneously. During the period where these transistors are ON, the transistor P**5** is turned ON to reset the gate potential of the driving transistor P**1** and then turned OFF. When the transistors P**3** and P**4** are ON, the transistor P**1** is a diode-connected transistor. A threshold compensation voltage based on the power supply potential VDD and the reference potential Vref is written to the storage capacitive element C**0**.
(42) The transistor P**2** is a switching transistor for selecting a pixel circuit to be supplied with a data signal and writing the data signal (data signal voltage) Vdata to the storage capacitive element C**0**. One end of the source/drain of the transistor P**2** is connected with the storage capacitive element C**0** and the other end is connected with a data line **237** for transmitting the data signal Vdata.
(43) The gate of the transistor P**2** is connected with a control signal line **231**C for transmitting a selection signal S**2** from the scanning driver **131**. The transistor P**2** is controlled by the selection signal S**2**. The selection signal S**2** is a signal different from the selection signal S**1**. For the pixel circuit **200**, the selection signal S**2** is a selection signal for controlling supply of the data signal Vdata to the storage capacitive element C**0**. When the transistor P**2** is ON, the transistor P**2** supplies the data signal Vdata supplied from the driver IC **134** through the data line **237** to the storage capacitive element C**0**.
(44) Structure of Pixel Unit
(45) Hereinafter, a structure of a pixel unit in the display region **125** is described. FIG. **3** is a plan diagram schematically illustrating a device structure of one pixel unit **300**. FIG. **3** schematically illustrates one entire pixel unit **300** and parts of neighboring pixel units. The pixel unit **300** is also referred to as pixel region. The pixel unit **300** has the circuit configuration described with reference to FIG. **2**. Specifically, the pixel unit **300** includes six thin-film transistors P**1** to P**6** and two capacitive elements C**1** and C**2** as illustrated in FIG. **3**. The light-emitting element E**1** is excluded from FIG. **3** for simplicity of illustration.
(46) FIG. **3** further includes a VDD power line **241** for transmitting the power supply potential VDD and a power line **242** for transmitting the reset potential Vrst. The reference potential Vref can be the same as the anode power supply potential VDD. Moreover, FIG. **3** includes a control signal lines **231**A and **231**B for transmitting the selection signal S**1**, a control signal line **231**C for transmitting the selection signal S**2**, and a control signal line **232**A for transmitting the emission control signal Em. FIG. **3** also includes a data line **237** for transmitting the data signal Vdata.
(47) In FIG. **3**, the metal layer (second metal layer to be described later) including the power line **241** is outlined by dashed lines. The parts of one same layer are denoted by the identical patterns. One layer is made of the same material by the same process. Each layer can be a single component layer made of a single material or can have a multi-layered structure consisting of a plurality of component layers made of different materials. In FIG. **3**, via regions (also referred to as contact regions), which are conductive regions provided in holes opened perpendicularly through an insulator to interconnect different layers, are denoted by filled squares or squares with an X inside.
(48) The squares with an X inside represent via regions between a third metal layer to be described

later and another conductive layer and the filled squares represent via regions between the second metal layer and another conductive layer. For example, the via region **301** interconnects the third metal layer and the VDD power line **241** and the via region **311** interconnects the third metal layer and the anode electrode of the OLED element E**1**.

(49) The third metal layer includes a main region **361** occupying a large area of the pixel unit **300** and an island region **362** provided inside an opening of the main region **361**. The main region **361** fills all area surrounding the opening and occupies more than a half of the area of the pixel unit **300**. The island region **362** is completely surrounded by a gap and physically and electrically isolated from the main region **361**.

(50) FIG. **4** schematically illustrates the cross-sectional structure along the section line IV-IV′ in FIG. **3**. FIG. **4** schematically illustrates the cross-sectional structure of a configuration of a pixel unit **300** including a flexible substrate, a driving TFT, and an OLED element of a TFT substrate **100** and a structural encapsulation unit **150**. In the following description, the definitions of upper side and lower side correspond to the upper side and the lower side of the drawing.

(51) The TFT substrate **100** includes a flexible substrate and pixel circuits (a TFT array) and OLED elements fabricated on the flexible substrate. The flexible substrate includes a polyimide layer **402**, a silicon oxide layer (SiOx layer) **403**, an amorphous silicon layer (a-Si layer) **404**, and a polyimide layer **405** in this order from the bottom. The silicon oxide layer **403** is formed above and in direct contact with the polyimide layer **402**. The amorphous silicon layer **404** is formed above and in direct contact with the silicon oxide layer **403**. The polyimide layer **405** is formed above and in direct contact with the amorphous silicon layer **404**.

(52) The TFT substrate **100** further includes a silicon oxide layer **406**, a transparent conductive layer **407**, a silicon oxide layer **408**, a silicon nitride layer (SiNx layer) **409**, and a silicon oxide layer **410** in this order from the bottom above the flexible substrate (polyimide layer **405**). The transparent conductive layer **407** can be made of an amorphous oxide (such as ITO or IZO) or amorphous silicon (a-Si). The pixel circuits (TFT array) and OLED elements are fabricated above the silicon oxide layer **410**.

(53) The silicon oxide layer **406** is formed between and in direct contact with the polyimide layer **405** and the transparent conductive layer **407**. The silicon oxide layer **406** fully covers the polyimide layer **405**. The silicon oxide layer **408** is formed above and in direct contact with the transparent conductive layer **407**. The silicon nitride layer **409** is formed above and in direct contact with the silicon oxide layer **408**. The silicon oxide layer **410** is formed above and in direct contact with the silicon nitride layer **409**.

(54) OLED elements are fabricated above the flexible substrate including the above-described multiple layers. Each OLED element includes a lower electrode region (for example, an anode electrode **438**), an upper electrode region (for example, a cathode electrode **432**), and a multilayer organic light-emitting film **434** located between the cathode electrode **432** and the anode electrode **438**. A plurality of anode electrodes **438** are disposed on the same plane (for example, on a planarization film **431**) and one multilayer organic light-emitting film **434** is disposed above one anode electrode **438**. In the example of FIG. **4**, the cathode electrode **432** of one pixel is a part of an unseparated conductive film covering the whole display region **125**.

(55) Illustrated in FIG. **4** is an example of a top-emission pixel structure, which includes top-emission type of OLED elements. The top-emission pixel structure is configured in such a manner that a cathode electrode **432** common to a plurality of pixels is provided on the light emission side (the upper side of the drawing). The cathode electrode **432** has a shape that fully covers the display region **125**. The top-emission pixel structure is characterized by that the anode electrode **438** has light reflectivity and the cathode electrode **432** has light transmissivity. Hence, a configuration to transmit light coming from the multilayer organic light-emitting film **434** toward the structural encapsulation unit **150** is attained. The cathode electrode **432** can be a semi-transparent film disposed at an optically optimum distance from the anode electrodes **438** to form a resonant

structure.

(56) The bottom-emission pixel structure has a transparent anode electrode and a reflective cathode electrode to emit light to the external through a flexible substrate. If both the anode electrode and the cathode electrode are made of light transmissive materials, a transparent display device can be obtained. The structure of the flexible substrate of this disclosure is applicable to OLED display devices of any of these types and further, display devices including light-emitting elements different from OLEDs.

(57) A subpixel of a full-color OLED display device usually lights in one of the colors of red, green, and blue. A red subpixel, a green subpixel, and a blue subpixel constitute one main pixel. A pixel circuit including a plurality of thin-film transistors controls light emission of an OLED element associated therewith. An OLED element is composed of an anode electrode of a lower electrode region, an organic light-emitting film, and a cathode electrode of an upper electrode region.

(58) Each pixel unit **300** includes a pixel circuit including a plurality of switching transistors and a driving transistor. The pixel circuit is fabricated between the silicon oxide layer **410** and an anode electrode **438** and controls the electric current to be supplied to the anode electrode **438**. The transistor P**3** illustrated in FIG. **4** has a top-gate structure. The other transistors also have a top-gate structure.

(59) A polysilicon layer is provided above and in direct contact with the silicon oxide layer **410**. The polysilicon layer includes a channel **413** that determines the transistor characteristics of the transistor P**3** at the location where a gate electrode region **417** is to be formed later. At both ends of the channel **413**, source/drain regions **414** and **415** are provided. The source/drain regions **414** and **415** are doped with high-concentration impurities to be electrically connected with a wiring layer thereabove. FIG. **4** further includes a source/drain region **412** of a transistor P**5**.

(60) Lightly doped drains (LDDs) doped with low-concentration impurities can be provided between the channel **413** and the source/drain region **414** and between the channel **413** and the source/drain region **415**. FIG. **4** omits the LDDs to avoid complexity. The polysilicon layer further includes regions to be used as parts of lines and electrodes.

(61) Above the polysilicon layer, a first metal layer including the gate electrode region **417** and electrode regions **418** and **419** is provided with a gate insulating layer **416** interposed therebetween. The electrode region **418** includes a gate electrode region of a not-shown driving transistor P**1**. The first metal layer can be made of a metal having a high melting point, such as W, Mo, or Ta, or an alloy of such a metal.

(62) An interlayer insulating film **420** is provided to cover the first metal layer. The interlayer insulating film **420** can be made of silicon oxide or silicon nitride. An intermediate conductive layer including electrode regions **421** and **422** is provided above and in direct contact with the interlayer insulating film **420**. The intermediate conductive layer can be made of a low-resistive semiconductor, a metal having a high melting point such as W, Mo, or Ta, or an alloy of such a metal. The intermediate conductive layer increases the capacitance of the storage capacitive element C**0** with a smaller area.

(63) In the example of FIG. **4**, the electrode region **421** overlaps the electrode region **419** of the first metal layer in the layering direction with the interlayer insulating film **420** interposed therebetween. The electrode region **422** overlaps the electrode region **418** of the first metal layer in the layering direction with the interlayer insulating film **420** interposed therebetween. A part of the first capacitive element C**1** is formed between the electrode regions **421** and **419**. The electrode region **419** is supplied with the anode power supply potential VDD. A part of the second capacitive element C**2** is formed between the electrode regions **422** and **418**.

(64) A planarization film **424** is provided to cover the intermediate conductive layer and the interlayer insulating film **420**. The planarization film **424** can be made of an organic or inorganic insulator and it is in direct contact with the intermediate conductive layer and the interlayer

insulating film **420**.
(65) A second metal layer including a source/drain electrode region **425** is provided above the planarization film **424**. The second metal layer can have a single layer structure of aluminum or a multilayered structure of Ti/Al/Ti. The source/drain electrode region **425** is connected with the source/drain region **414** of the polysilicon layer by a via region provided in a hole opened in the layering direction through the interlayer insulating film **420** and the gate insulating layer **416**.
(66) The second metal layer further includes an electrode region **426**. The electrode region **426** is connected with the electrode region **422** of the intermediate conductive layer by a via region provided in a hole opened in the layering direction through the planarization film **424**. These electrode regions are at the same potential.
(67) The second metal layer still further includes an electrode region **427**. The electrode region **427** is opposed to the electrode region **421** of the intermediate conductive layer in the layering direction and the interlayer insulating film **420** is interposed therebetween. A part of the first capacitive element C**1** is formed between the electrode regions **427** and **421**.
(68) The second metal layer still further includes an electrode region **428**. The electrode region **428** is connected with the source/drain region **412** of the transistor P**5** by a via region provided in a hole opened straight downward through the planarization film **424**, the interlayer insulating film **420**, and the gate insulating layer **416**.
(69) Another planarization film **430** is provided to cover the second metal layer and the planarization film **424**. The planarization film **430** can be made of an organic or inorganic insulator and it is in direct contact with the second metal layer and the planarization film **424**.
(70) A third metal layer including the main region **361** and the island region **362** is provided above the planarization film **430**. The third metal layer can have a single layer structure of aluminum or a multilayered structure of Ti/Al/Ti; it is in direct contact with the planarization film **430**. As described above, the island region **362** is isolated from the main region **361** and surrounded by the main region **361** with a gap. The main region **361** surrounds the island region **362** and there is a gap around the island region **362**.
(71) The main region **361** is connected with the electrode region **427** of the second metal layer by a via region **301** extending straight downward in a hole of the planarization film **430**. The main region **361** is supplied with the anode power supply potential VDD through the electrode region **427**.
(72) The main region **361** in this example covers the entire intermediate conductive layer when viewed in the layering direction (in the vertical direction in FIG. **4**). The main region **361** is partially opposed to parts of the electrode regions **421** and **422** of the intermediate conductive layer across the planarization films **430** and **424**. The main region **361** forms a part of the first capacitive element C**1** with the electrode regions **421** and **422**. Furthermore, the main region **361** is partially opposed to the electrode region **426** of the second metal layer across the planarization film **430**. The main region **361** forms another part of the first capacitive element C**1** with the electrode region **426**.
(73) As described above, the first capacitive element C**1** in this example includes a capacitor between the electrode region **421** of the intermediate conductive layer and the electrode region **419** of the first metal layer, a capacitor between the electrode region **421** and the electrode region **427** of the second metal layer, a capacitor between the electrode regions **421** and **422** of the intermediate conductive layer and the main region **361** of the third metal layer, and a capacitor between the electrode region **426** of the second metal layer and the main region **361** of the third metal layer.
(74) The electrode region **427** of the second metal layer and the main region **361** of the third metal layer are at the same potential. The electrode region **426** of the second metal layer and the electrode regions **421** and **422** of the intermediate conductive layer are at the same potential. The electrode regions **419** and **427** and the main region **361** are supplied with the anode power supply potential

VDD. As understood from this description, the first capacitive element C**1** includes a capacitor between the intermediate conductive layer and the third metal layer, in addition to a capacitor between the intermediate conductive layer and the first metal layer.
(75) The second capacitive element C**2** consists of a capacitor between the electrode region **422** of the intermediate conductive layer and the electrode region **418** of the first metal layer. The electrode region **418** extends to the gate electrode of the driving transistor P**1**. As understood from this description, the second capacitive element C**2** consists of only the capacitor between the intermediate conductive layer and the first metal layer.
(76) As described above, the intermediate conductive layer forms a capacitor included in the second capacitive element C**2** with the first metal layer and forms capacitors included in the first capacitive element C**1** with the main region **361**, the second metal layer, and the first metal layer.
(77) The main region **361** of the third metal layer occupies a large area of the pixel unit **300** and covers the whole intermediate conductive layer including the electrode regions **421** and **422** of the storage capacitive element C**0**. Since the main region **361** is supplied with a constant anode power supply potential VDD, the voltage variation in the storage capacitive element C**0** caused by crosstalk can be reduced. Furthermore, the power line attains a larger area with the main region **361**, which reduces the possibility of breakage of the power line caused by bend and the IR drop.
(78) The island region **362** is connected with the electrode region **428** of the second metal layer by a via region **312** under the island region **362**. The via region **312** extends in the layering direction through the planarization film **430** and is in direct contact with the electrode region **428**.
(79) A planarization film **431** is provided to cover the third metal layer and the planarization film **430**. The planarization film **431** can be made of an organic or inorganic insulator and is in direct contact with the third metal layer and the planarization film **430**.
(80) An anode electrode **438** is provided above the planarization film **431**. The anode electrode **438** is in direct contact with the planarization film **431**. The anode electrode **438** is connected with the source/drain region **412** through a contact hole of the planarization film **431**. The TFTs of the pixel circuit are fabricated under the anode electrode **438**. The anode electrode **438** consists of a reflective metal layer in the middle and transparent conductive layers sandwiching the reflective metal layer, for example. The anode electrode **438** can have a structure of ITO/Ag/ITO or IZO/Ag/IZO.
(81) The anode electrode **438** is connected with the island region **362** of the third metal layer by a via region under the anode electrode **438**. The via region extends in the layering direction through the planarization film **431** and is in direct contact with the island region **362**. The source/drain region **412**, the electrode region **428** of the second metal layer, and the island region **362** supply a driving potential (anode potential) that determines the brightness (the amount of light) of the light-emitting element. The island region **362** is distant from and outside the main region **361** to be supplied with the power supply potential VDD. For this reason, the island region **362** can supply the driving potential to the anode electrode **438**.
(82) An insulating pixel defining layer (PDL) **437** is provided above the anode electrode **438** to separate OLED elements. Each OLED element is fabricated in an opening **436** of the pixel defining layer **437**.
(83) Above the anode electrode **438**, a multilayer organic light-emitting film **434** is provided. The multilayer organic light-emitting film **434** is in contact with the pixel defining layer **437** in an opening **436** of the pixel defining layer **437** and its periphery. The multilayer organic light-emitting film **434** is formed by depositing organic light-emitting material for the color of R, G, or B on the anode electrode **438**.
(84) The multilayer organic light-emitting film **434** is formed by vapor deposition of organic light-emitting material in the region corresponding to a pixel through a metal mask. The multilayer organic light-emitting film **434** consists of, for example, a hole injection layer, a hole transport layer, a light-emitting layer, an electron transport layer, and an electron injection layer in this order

from the bottom. The layered structure of the multilayer organic light-emitting film **434** is determined depending on the design.
(85) A cathode electrode **432** is provided over the multilayer organic light-emitting film **434**. The cathode electrode **432** is a light-transmissive electrode. The cathode electrode **432** transmits part of the visible light coming from the multilayer organic light-emitting film **434**. The layer of the cathode electrode **432** is formed by vapor deposition of a metal such as Ag or Mg or an alloy thereof. If the resistance of the cathode electrode **432** is so high to impair the uniformity of the brightness of emitted light, an additional auxiliary electrode layer is provided using a material for a transparent electrode, such as ITO, IZO, ZnO, or In.sub.2O.sub.3.
(86) The stack of an anode electrode **438**, a multilayer organic light-emitting film **434**, and a cathode electrode **432** formed in an opening **436** of the pixel defining layer **437** corresponds to an OLED element. A structural encapsulation unit **150** is provided above and in direct contact with the cathode electrode **432**. The structural encapsulation unit (thin-film encapsulation unit) **150** includes an inorganic insulator (for example, SiNx or AlOx) layer **433**, an organic planarization film **439**, and another inorganic insulator (for example, SiNx or AlOx) layer **435**.
(87) A touch screen film **453**, a λ/4 plate **454**, a polarizing plate **455**, and a resin cover lens **456** are laid in this order from the bottom, on the encapsulation structural unit **150**. The λ/4 plate **454** and the polarizing plate **455** are to reduce the reflection of the light coming from the external. The layered structure of the OLED display device described with reference to FIG. **4** is an example; one or more of the layers in FIG. **4** may be omitted and one or more layers not shown in FIG. **4** may be added. For example, the intermediate conductive layer can be excluded. If the intermediate conductive layer is excluded, a storage capacitor is formed between the polysilicon layer doped with high-concentration impurities and the gate electrode with the gate insulating layer **416** interposed therebetween.
(88) FIGS. **5**A to **5**G illustrate patterns of some layers included in the layered structure of the TFT substrate **100**. FIGS. **5**A to **5**G are sorted to illustrate the layers in order of formation on the flexible substrate (from the bottom).
(89) FIG. **5**A schematically illustrates a semiconductor (such as polysilicon) layer pattern **501**. The semiconductor layer pattern **501** includes channel regions having high resistance. Some parts of the semiconductor layer pattern **501** are doped with impurities to become low-resistive regions having lower resistance than the channel regions.
(90) FIG. **5**B schematically illustrates a first metal layer pattern **503**. As described above, the first metal layer includes gate electrode regions for the transistors and further, line regions and electrode regions for transmitting a power supply potential or a signal potential. FIG. **5**C schematically illustrates an intermediate conductive layer pattern **505**. The intermediate conductive layer includes electrode regions **421** and **422** and a connector region **423** for connecting these electrode regions. The electrode regions **421** and **422** are a fifth electrode region and a fourth electrode region, respectively.
(91) The width (the length in the horizontal direction in FIG. **5**C) of the connector region **423** is smaller than the width of the electrode regions **421** and **422**. It is preferable that the width of the connector region **423** be as smaller as possible to minimize the effect of misalignment or variation in processing onto the two storage capacitors and accordingly, the minimum value allowed for the manufacturing process is employed. For example, it can be 1/10 or less of the width of the electrode regions **421** and **422**. As described above, the intermediate conductive layer is a component of the storage capacitors.
(92) FIG. **5**D schematically illustrates the layered structure of the semiconductor layer pattern **501**, the first metal layer pattern **503**, and the intermediate conductive layer pattern **505**. FIG. **5**D further includes via regions for interconnecting the second metal layer and a conductive layer lower than the second metal layer. The via regions are denoted by filled squares. The via region **531** interconnects the first metal layer and the second metal layer and the via region **551** interconnects

the intermediate conductive layer and the second metal layer.
(93) FIG. **5**E schematically illustrates a second metal layer pattern **507**. FIG. **5**E further includes via regions **301** and **312** for interconnecting the second metal layer and the third metal layer. As described above, the via region **301** interconnects the main region **361** of the third metal layer and the second metal layer. The via region **312** interconnects the island region **362** of the third metal layer and the second metal layer. FIG. **5**F schematically illustrates a third metal layer pattern **509**. FIG. **5**F further includes a via region **311** for interconnecting the third metal layer and the anode electrode of the OLED element E**1**. As illustrated in FIG. **5**F, the main region **361** fills all area of the pixel unit except for the island region **362** and the gap between the main region **361** and the island region **362**.
(94) The main region **361** of one pixel unit **300** is a part of one unseparated metal sheet. This metal sheet includes the main regions **361** of a plurality of pixel units **300**. As to the island region **362**, one pixel unit **300** has one island region **362**. The island region **362** is located inside an opening **363** of the main region **361** and the outer end of the island region **362** is distant from the inner end of the opening **363** of the main region **361** at any point.
(95) FIG. **5**G schematically illustrates the anode electrode **438** and an opening **436** of the pixel defining layer **437**. The light-emitting region is located inside the opening **436**. As described above, the driving potential is supplied to the anode electrode **438** through the island region **362** of the third metal layer.
(96) Configuration of Display Region
(97) Hereinafter, a configuration example of the display region **125** in an embodiment of this specification is described. FIG. **6** schematically illustrates a configuration example of a display region **125**. The OLED display device **10** can be mounted on a mobile terminal, such as a smartphone or a tablet terminal. The display region **125** includes a normal region **461** having a normal pixel density and a low-density region **463** of N columns and M rows having a pixel density lower than the pixel density of the normal region **461**. The normal region **461** is a first display region and the low-density region **463** is a second display region.
(98) One or more cameras **465** are disposed under (behind) the low-density region **463**. A camera **465** is an example of photosensor; other kinds of photosensors can be disposed under the low-density region **463**. In FIG. **6**, one of a plurality of cameras is provided with a reference sign **465** by way of example.
(99) When viewing the display region **125**, the cameras **465** are located behind the low-density region **463**; each camera **465** takes a picture of an object in front of the camera with light transmitted through the low-density region **463**. Not to interfere with the cameras **465** taking a picture, the density of pixel units (including a light-emitting element and a pixel circuit) in the low-density region **463** is lower than the density of pixel units in the surrounding normal region **461**.
(100) A not-shown controller sends the data of the pictures taken by the cameras **465** to the OLED display device **10**. Although the example of a low-density region in FIG. **6** is a region where cameras are disposed thereunder, the features of this description are applicable to display devices including a region having a relatively low pixel density for other purposes. The frequency of the light to be sensed by the photosensor is not limited to a specific value.
(101) Layout of Lines and Third Metal Layer
(102) Hereinafter, an example of the wiring layout of the OLED display device **10** is described. FIG. **7** schematically illustrates a layout of control lines on the TFT substrate **100**. The low-density region **463** in the configuration example in FIG. **7** includes an end of the display region **125**; one side of the low-density region **463** is a part of one side of the display region **125**. The remaining outer end of the low-density region **463** is located inside the display region **125** and it is the boundary with the normal region **461**.
(103) The layout of pixel circuits in the normal region **461** in the configuration example of FIG. **7** is stripe arrangement. Specifically, each pixel column (subpixel column) extending along the Y-axis

is composed of pixels (subpixels) of the same color. Each pixel row extending along the X-axis is composed of red pixels, green pixels, and blue pixels disposed cyclically. The low-density region **463** has a configuration such that some pixels are removed from the pixel layout of the normal region **461**. The blank area in the low-density region **463** does not include any pixel circuit including an OLED element but includes only lines.

(104) A plurality of scanning lines **106** extend along the X-axis from the scanning driver **131**. A plurality of emission control lines **107** extend along the X-axis from the emission driver **132**. In FIG. **7**, one of the scanning lines and one of the emission control lines are provided with reference signs **106** and **107**, respectively. A scanning line **106** in the configuration example of FIG. **7** transmits a selection signal for pixels in the normal region **461** and the low-density region **463**. An emission control line **107** transmits an emission control signal for the normal region **461** and the low-density region **463**.

(105) The driver IC **134** sends a control signal for the scanning driver **131** through lines **711** and a control signal for the emission driver **132** through lines **713**. The driver IC **134** controls the timing of the scanning signal (selection pulses) from the scanning driver **131** and the emission control signal from the emission driver **132**, based on image data (an image signal) from the external.

(106) The driver IC **134** supplies data signals for the subpixels in the normal region **461** and the low-density region **463** to the demultiplexer **136** through lines **705**. In FIG. **7**, one of the lines is provided with a reference sign **705** by way of example. The driver IC **134** determines data signals for individual subpixels in the normal region **461** and the low-density region **463**. The data signal for one subpixel is determined from grayscale levels of one or more subpixels of image data (for one frame) from the external.

(107) The demultiplexer **136** outputs one output of the driver IC **134** to N data lines (N is an integer greater than 1) in series within a scanning period. In FIG. **7**, one of the plurality of data lines extending along the Y-axis is provided with a reference sign **105**, by way of example.

(108) FIG. **7** also schematically illustrates a layout of an anode power line pattern on the TFT substrate **100**. As illustrated in FIG. **7**, the TFT substrate **100** includes an anode power line pattern **801**. The anode power line pattern **801** supplies an anode power supply potential to the pixel circuits in the normal region **461** and the low-density region **463**.

(109) The anode power line pattern **801** includes a rectangular surrounding part and a line part inside the surrounding part. In the line part, lines extending along the Y-axis are disposed side by side along the X-axis. Some lines extend through the normal region **461** and the low-density region **463**. The anode power line pattern **801** transmits the anode power supply potential VDD to the pixel circuits of individual subpixels in the normal region **461** and the low-density region **463**. The anode power line pattern **801** can be different from the example of FIG. **7**. For example, the anode power line pattern **801** can be spread like a mesh including a plurality of lines extending along the X-axis, in addition to the lines extending along the Y-axis.

(110) FIG. **7** further schematically illustrates the third metal layer **591**. The main regions of the third metal layer **591** are supplied with the anode power supply potential VDD from the anode power line pattern **801**. In this example, the third metal layer **591** is provided in the normal region **461** and not provided in the low-density region **463**. Since the third metal layer **591** is removed from (is not provided in) the low-density region **463** and is provided only outside the low-density region **463**, the reduction of the light that passes through the low-density region **463** to reach the cameras **465** behind the low-density region **463** can be made small.

(111) Unlike the configuration where the low-density region **463** is located inside an opening of the third metal layer, the third metal layer **591** can be partially located in the low-density region **463**. The third metal layer **591** in the low-density region **463** has a different pattern from the pattern of the third metal layer **591** in the normal region **461**.

(112) The occupancy of the third metal layer **591** in the low-density region **463** is lower than the one in the normal region **461**. This configuration reduces the reduction of light to reach the cameras

465. The occupancy in a region is a value obtained by dividing the area of the region filled with the third metal layer **591** by the area of the whole region.
(113) In an example, the pixel circuits in the normal region **461**, the pixel circuits in the low-density region **463**, and the pixel units have the same external dimensions. The pixel units in the normal region **461** have a structure described with reference to FIGS. **3** to **5**G.
(114) The low-density region **463**, however, does not include the third metal layer **591** and accordingly, the pixel units in the low-density region **463** have a structure such that the third metal layer is removed from the structure described with reference to FIGS. **3** to **5**G. For this reason, the via region **311** shown in FIG. **4** is in direct contact with the electrode region **428** of the second metal layer. In other words, the anode electrode **438** is connected with the second metal layer by a via region extending straight downward, not via the third metal layer. The pixel units in the low-density region **463** may include the third metal layer. For example, the third metal layer can have only the island pattern at the location of the via region **311**. This configuration equalizes contact resistances from a driving TFT to an anode electrode between the low-density region and the high-density region (normal region). The occupancy of the third metal layer in a pixel unit (pixel region) in the low-density region **463** is lower than the one in the normal region **461**.
(115) The driver IC **134** includes a DC-DC converter to generate a plurality of power supply potentials and supplies the power supply potentials to the OLED display panel. The driver IC **134** outputs the anode power supply potential VDD to the anode power line pattern **801** and outputs the cathode power supply potential VSS to the cathode electrode **432**.
(116) The cathode electrode layer not shown in FIG. **7** has a shape of one sheet and fully covers the normal region **461** and the low-density region **463**. The cathode electrode of each pixel in these regions **461** and **463** is a part of this one sheet of cathode electrode layer.
OTHER CONFIGURATION EXAMPLES
(117) Hereinafter, another structure of a pixel unit in the normal display region **461** is described. In the structural example described with reference to FIGS. **3** to **5**G, the main region of the third metal layer covers the whole intermediate conductive layer in a pixel unit when viewed in the layering direction. In another configuration example of a pixel unit, the main region covers only a part of the intermediate conductive layer. The capacitance of the first capacitive element C**1** can be adjusted by adjusting the area of the main region covering the intermediate conductive layer.
(118) FIG. **8**A illustrates a first metal layer pattern **503**, an intermediate conductive layer pattern **505**, and a third metal layer pattern **820** in another structural example of a pixel unit including a third metal layer. Compared to the structure described with reference to FIGS. **3** to **5**G, the first metal layer pattern **503** and the intermediate conductive layer pattern **505** are the same and the third metal layer pattern **820** is different.
(119) FIG. **8**B illustrates the intermediate conductive layer pattern **505** and the third metal layer pattern **820** in the structural example illustrated in FIG. **8**A. The third metal layer pattern **820** in a pixel unit consists of a main region **821** and an island region **822**. The island region **822** is located inside an opening of the main region **821**.
(120) There is a gap around the island region **822** to isolate the island region **822** from the main region **821**. This gap has a larger area than that of the foregoing structural example. Accordingly, the occupancy of the third metal layer in the pixel unit is lower than that of the foregoing structural example. The electrode region **421** of the intermediate conductive layer is located within the gap between the main region **821** and the island region **822**; it does not overlap the third metal layer pattern **820** in the layering direction. However, the whole electrode region **422** of the intermediate conductive layer overlaps (is covered with) the third metal layer pattern **820** in the layering direction.
(121) The intermediate conductive layer pattern **505** includes a connector region **423** connecting the two electrode regions **421** and **422** in the Y-axis direction (first direction). It is preferable that the width (the length in the X-axis direction (second direction) in FIG. **8**B) of the connector region

423 be smaller than the width of the electrode regions **421** and **422**; the minimum value allowed for the manufacturing process is employed. As a result, the variation in capacitance of the storage capacitor caused by an alignment error in the Y-axis direction of the intermediate conductive layer can be minimized.

(122) As illustrated in FIG. **8**B, distances L**1** and L**2** are defined between an end of the electrode region **421** of the intermediate conductive layer and an end (inner end of the opening) of the main region **821** of the third metal layer. In an example, these distances L**1** and L**2** are equal. This configuration reduces the effect of alignment error.

(123) FIG. **8**C illustrates a second metal layer pattern **850**, in addition to the layered structure illustrated in FIG. **8**A. Since this embodiment is configured so that the third metal layer does not overlap the electrode region **421** of the intermediate conductive layer, a via region **825** for connecting the second metal layer pattern **850** and the third metal layer pattern **820** is provided at a different location, compared to the foregoing structural example. The via region **825** interconnects the main region **821** of the third metal layer and an electrode region (line region) of the second metal layer for transmitting the anode power supply potential VDD, like the via region **301** in the foregoing structural example.

(124) FIG. **9** is a cross-sectional diagram schematically illustrating the cross-sectional structure of this structural example of a pixel unit. The following mainly describes differences from the structural example in FIG. **4**. The electrode region **421** of the intermediate conductive layer is not covered with the third metal layer pattern and is located inside an opening. Since the via region connecting the second metal layer and the third metal layer is provided at a different location, this example does not include the via region **301** and the electrode region **427** of the second metal layer in FIG. **4**.

(125) Like in the structural example in FIG. **4**, a part of the first capacitive element C**1** is formed between the main region **821** of the third metal layer and the electrode region **426** of the second metal layer and another part of the first capacitive element C**1** is formed between the main region **821** and the electrode region **422** of the intermediate conductive layer. The second capacitive element C**2** is formed only between the intermediate conductive layer and the first metal layer.

(126) Another example of application of the third metal layer is described. The example described in the following uses the third metal layer as a pinhole matrix layer. The pinhole matrix layer includes a pinhole array including a plurality of pinholes disposed in two dimensions. A pinhole transmits only the light coming in a specific direction from a point of an object.

(127) FIG. **10** is a cross-sectional diagram schematically illustrating the cross-sectional structure of a low-density region **463**. In the low-density region **463**, the third metal layer functions as a matrix pinhole image sensing (MAPIS) layer **593**. The MAPIS layer **593** is a pinhole matrix layer.

(128) A structural encapsulation unit **471** is laid to cover OLED elements E**1**. A touch screen film **453**, a polarizing plate **455**, and a cover lens **456** are laid in this order from the bottom above the structural encapsulation unit **471**. The third metal layer pattern in the normal region **461** is the same as the one described with reference to FIGS. **3** to **5**G or FIGS. **8**A to **9**. As understood from this description, the pattern of the third metal layer in the low-density region **463** is different from the pattern of the third metal layer in the normal region **461**.

(129) In FIG. **10**, a CMOS sensor **901** of a photosensor is disposed under (behind) the MAPIS layer **593**. The configuration example illustrated in FIG. **10** can function as a fingerprint sensor, for example. A finger **902** to be imaged reflects light from the OLED elements E**1**. The light from the finger **902** passes through the pinhole array of the MAPIS layer **593** to form an image on the CMOS sensor **901**. The image of the fingerprint taken by the CMOS sensor **901** is transferred to a not-shown controller. The controller compares the acquired fingerprint image with the fingerprint data of a registered user to authenticate the user.

(130) The MAPIS layer **593** is supplied with the anode power supply potential VDD. The MAPIS layer **593** partially forms one or more capacitors included in a storage capacitive element. All area

of the MAPIS layer **593** except for the pinholes is filled with metal. In an example, the occupancy of the MAPIS layer **593** in a pixel unit is higher than the occupancy of the third metal layer in a pixel unit in the normal region **461**. As illustrated in this configuration example and other configuration examples, the third metal layer having different patterns in different regions of the display regions **125** can provide the photosensor behind the third metal layer with light appropriate for the purpose of the region.
(131) As set forth above, embodiments of this disclosure have been described; however, this disclosure is not limited to the foregoing embodiments. Those skilled in the art can easily modify, add, or convert each element in the foregoing embodiments within the scope of this disclosure. A part of the configuration of one embodiment can be replaced with a configuration of another embodiment or a configuration of an embodiment can be incorporated into a configuration of another embodiment.

Claims

1. A display device comprising: a first display region configured to display an image; a second display region configured to display an image, the second display region having a lower pixel density than the first display region; a first metal layer; a second metal layer located upper than the first metal layer; and a third metal layer located upper than the first metal layer and the second metal layer, wherein occupancy of the third metal layer in the second display region is lower than occupancy of the third metal layer in the first display region, wherein the first display region includes a plurality of first pixel units and each of the plurality of first pixel units includes: a light-emitting element including an upper electrode region, a lower electrode region, and a light-emitting layer located between the upper electrode region and the lower electrode region; and a driving transistor configured to control light emission of the light-emitting element, wherein the first metal layer, the second metal layer, and the third metal layer are located lower than the lower electrode region, wherein, in each of the plurality of first pixel units, the first metal layer includes a first electrode region configured to control an amount of electric current in a channel of the driving transistor, wherein, in each of the plurality of first pixel units, the second metal layer includes a second electrode region and a third electrode region configured to supply electric current to the channel of the driving transistor, and wherein, in each of the plurality of first pixel units, the third metal layer includes: a main region configured to be supplied with a power supply potential and form a capacitor included in a first capacitive element with the second metal layer, the first capacitive element being configured to store a voltage to control the driving transistor; and an island region separated from the main region and surrounded by the main region with a gap, the island region being interconnected with the lower electrode region by a via region.
2. The display device according to claim 1, further comprising an intermediate conductive layer between the first metal layer and the second metal layer, wherein, in each of the plurality of first pixel units, the intermediate conductive layer is configured to form a capacitor included in a second capacitive element different from the first capacitive element with the first metal layer, and wherein, in each of the plurality of first pixel units, the intermediate conductive layer is configured to form a capacitor included in the first capacitive element with the main region.
3. The display device according to claim 2, wherein, in each of the plurality of first pixel units, the main region covers all area of the intermediate conductive layer when viewed in a layering direction.
4. The display device according to claim 1, wherein the main region fills all area of the first pixel unit except for the island region and the gap.
5. The display device according to claim 1, wherein the second display region does not include the third metal layer.
6. An electronic device comprising: the display device according to claim 1; and a photosensor

disposed under the second display region.

7. A display device comprising: a first display region configured to display an image; a second display region configured to display an image, the second display region having a lower pixel density than the first display region; a first metal layer; a second metal layer located upper than the first metal layer; and a third metal layer located upper than the first metal layer and the second metal layer, wherein a pattern of the third metal layer in the second display region is different from a pattern of the third metal layer in the first display region, wherein the first display region includes a plurality of first pixel units and each of the plurality of first pixel units includes: a light-emitting element including an upper electrode region, a lower electrode region, and a light-emitting layer located between the upper electrode region and the lower electrode region; and a driving transistor configured to control light emission of the light-emitting element, wherein the first metal layer, the second metal layer, and the third metal layer are located lower than the lower electrode region, wherein, in each of the plurality of first pixel units, the first metal layer includes a first electrode region configured to control an amount of electric current in a channel of the driving transistor, wherein, in each of the plurality of first pixel units, the second metal layer includes a second electrode region and a third electrode region configured to supply electric current to the channel of the driving transistor, and wherein, in each of the plurality of first pixel units, the third metal layer includes: a main region configured to be supplied with a power supply potential and form a capacitor included in a first capacitive element with the second metal layer, the first capacitive element being configured to store a voltage to control the driving transistor; and an island region separated from the main region and surrounded by the main region with a gap, the island region being interconnected with the lower electrode region by a via region.

8. The display device according to claim 7, wherein the third metal layer includes a pinhole array in the second display region.

9. An electronic device comprising: the display device according to claim 7; and a photosensor disposed under the second display region.

10. The display device according to claim 2, wherein, in each of the plurality of first pixel units, the intermediate conductive layer includes a fourth electrode region, a fifth electrode region, and a connector region connecting the fourth electrode region and the fifth electrode region in a first direction, wherein the whole fourth electrode region is covered with the third metal layer when viewed in a layering direction, wherein the fifth electrode region is located in a gap between the main region and the island region when viewed in the layering direction, and wherein the connector region has a smaller length than the fourth electrode region and the fifth electrode region in a second direction perpendicular to the first direction.

11. The display device according to claim 2, wherein, in each of the plurality of first pixel units, the intermediate conductive layer includes a fourth electrode region and a fifth electrode region, wherein the whole fourth electrode region is covered with the third metal layer when viewed in a layering direction, wherein the fifth electrode region is located in a gap between the main region and the island region when viewed in the layering direction, and wherein distances from two sides of the fifth electrode region to the main region are equal.